

ORIGINAL ARTICLE

Angiography-Based Physiology to Guide Coronary Revascularization

Joost Daemen, M.D., Ph.D.,¹ Jari A. van der Eijk, M.Sc.,¹ Marco Barbierato, M.D.,² Robert A. Byrne, M.B., B.Ch., Ph.D.,³⁻⁵ Paolo Canova, M.D.,⁶ Giovanni L. De Maria, M.D., Ph.D.,⁷ Eric Van Belle, M.D., Ph.D.,^{8,9} Ignacio Amat-Santos, M.D.,¹⁰ Antonio Mangieri, M.D.,¹¹ Thomas Hovasse, M.D., Ph.D.,¹² Manel Sabaté, M.D., Ph.D.,¹³ Ronan Margey, B.Med.Sci., M.B.,¹⁴ Simone Biscaglia, M.D., Ph.D.,¹⁵ Fernando Rivero, M.D., Ph.D.,¹⁶ Domenico D'Amario, M.D., Ph.D.,^{17,18} Youssef S. Abdelwahed, M.D., Ph.D.,^{19,20} Rohit M. Oemrawsingh, M.D., Ph.D.,²¹ Raul Moreno, M.D., Ph.D.,²² James Cockburn, M.B., B.S., M.D.,²³ Luca Testa, M.D., Ph.D.,²⁴ Ton de Vries, M.Sc.,²⁵ Ulf Landmesser, M.D., Ph.D.,¹⁹ Adrian P. Banning, M.D., Ph.D.,^{7,26} Jan G.P. Tijssen, Ph.D.,²⁷ and Ernest Spitzer, M.D.,^{25,28} for the FAST III Investigators*

ABSTRACT

BACKGROUND

Current society guidelines recommend physiological assessment of intermediate coronary lesions to guide revascularization. Data regarding revascularization guided by vessel fractional flow reserve (vFFR), derived from three-dimensional quantitative coronary angiography without the need for a pressure wire or hyperemic agent, as compared with pressure-wire–based fractional flow reserve (FFR), are lacking.

METHODS

We conducted an international, open-label, randomized, noninferiority trial at 37 sites in Europe. Patients with intermediate coronary-artery lesions (diameter stenosis of 30 to 80%) who presented with chronic or acute coronary syndromes were randomly assigned in a 1:1 ratio to undergo either vFFR-guided or FFR-guided revascularization of the intermediate coronary-artery lesions. The primary end point was a composite of death from any cause, any myocardial infarction, or any revascularization at 1 year. The noninferiority margin was 3.0 percentage points.

RESULTS

The primary end point was assessed in 1116 patients in the vFFR group and 1095 in the FFR group. The mean age of the patients was 67 years, 24.3% were women, 18.7% presented with an acute coronary syndrome, and 26.6% had diabetes mellitus. At 1 year, a primary end-point event had occurred in 80 patients (Kaplan–Meier estimate, 7.5%) in the vFFR group and in 79 patients (Kaplan–Meier estimate, 7.5%) in the FFR group (risk difference, –0.02 percentage points; 95% confidence interval, –2.25 to 2.21; $P=0.004$ for noninferiority). The incidence of serious adverse events appeared to be similar in the two groups.

CONCLUSIONS

Among patients with intermediate coronary lesions, vFFR-guided revascularization was noninferior to FFR-guided revascularization with respect to a composite of death, myocardial infarction, or revascularization at 1 year. (Funded by Pie Medical Imaging and Siemens Healthineers; FAST III ClinicalTrials.gov number, NCT04931771.)

Author affiliations are listed at the end of the article. Joost Daemen can be contacted at j.daemen@erasmusmc.nl or at the Department of Cardiology, Thoraxcenter, Erasmus University Medical Center, Dr. Molewaterplein 40, Rm. Rg-628, 3015 GD, Rotterdam, the Netherlands.

*A complete list of the FAST III investigators is provided in the Supplementary Appendix, available at NEJM.org.

This article was published on March 29, 2026, at NEJM.org.

N Engl J Med 2026;395:20-31.

DOI: 10.1056/NEJMoa2601841

Copyright © 2026 Massachusetts Medical Society.

GUIDELINES FOR THE MANAGEMENT OF coronary artery disease recommend the use of fractional flow reserve (FFR) to guide revascularization of intermediate coronary-artery lesions in the absence of noninvasive ischemia testing.^{1,3} With the intention to increase adoption of physiological guidance and reduce procedure time by avoiding intracoronary instrumentation and adenosine-induced hyperemia, angiography-based indexes were developed.⁴

Vessel fractional flow reserve (vFFR) (Pie Medical Imaging) derives an FFR value from angiographic images and three-dimensional quantitative coronary angiography by using physical laws that apply to coronary flow behavior, including viscous resistance and effects of separation loss, as described by Gould et al. and Kirkeeide.^{5,6} In prospective, multicenter validation studies, vFFR has shown diagnostic accuracy similar to that of pressure-wire–based FFR.^{7,9}

The clinical validation of vFFR occurred in parallel with other angiography-based indexes, with quantitative flow reserve being the most widely validated index. After successful validation, a multicenter trial showed that quantitative flow reserve–guided revascularization improved 1-year outcomes as compared with angiographically guided revascularization.¹⁰ On the basis of the results of that trial, quantitative flow reserve was given a class IB recommendation in contemporary guidelines.³ Conversely, the results of a more recent multicenter European trial showed higher rates of adverse cardiac events with quantitative flow reserve–guided revascularization than with FFR-guided revascularization.¹¹

The Fractional Flow Reserve or 3D-Quantitative-Coronary-Angiography–Based Vessel-FFR–Guided Revascularization (FAST III) trial was designed to assess whether vFFR-guided revascularization is noninferior to FFR-guided revascularization with respect to death from any cause, myocardial infarction, or revascularization.

METHODS

TRIAL DESIGN AND OVERSIGHT

We conducted an investigator-initiated, international, open-label, randomized, controlled, non-inferiority trial. The design of the trial has been published previously,¹² and the protocol is available with the full text of this article at NEJM.org. The trial was designed by the principal investiga-

tor and the executive committee and was approved by the ethics committee and corresponding health authorities at each participating site. All the patients provided written informed consent. The regulatory sponsor was the European Cardiovascular Research Institute, which received unrestricted research grants from Pie Medical Imaging and Siemens Healthineers. Pie Medical Imaging provided free software licenses and training throughout the trial; additional information is provided in the Supplementary Appendix, available at NEJM.org. The funders had no role in the design of the trial; the collection, monitoring, analysis, or interpretation of the data; or the preparation of the manuscript. A contract academic research organization (Cardialysis) managed data collection through a secure Web-based electronic case-report form system and performed all clinical trial and core laboratory activities, including statistical reporting. An independent data and safety monitoring board periodically reviewed all reported serious adverse events. An independent clinical-events committee adjudicated all components of the primary end point and other clinical events. The trial was conducted in accordance with the Declaration of Helsinki and standard ISO 14155 of the International Organization for Standardization (Clinical Investigation of Medical Devices for Human Subjects — Good Clinical Practice). The first author and the last author oversaw the conduct of the trial and drafted the manuscript. All the authors vouch for the accuracy and completeness of the data and for the fidelity of the trial to the protocol. No agreements existed with any funding source or third party that would have limited full publication of the findings.

PATIENT SELECTION

Eligible patients were 18 years of age or older and had chronic coronary syndrome, unstable angina, or non–ST-segment elevation acute coronary syndrome and at least one intermediate coronary-artery lesion (diameter stenosis of 30 to 80%, as determined by visual estimation or quantitative coronary angiography) in a native epicardial vessel with a reference diameter of 2.5 mm or greater that was deemed to be suitable for physiological assessment with both vFFR and wire-based FFR. Intermediate lesions had to be located in vessels with flow grade 3 according to the Thrombolysis in Myocardial Infarction flow grading system (grades range from 0 to 3, with higher grades



**A Quick Take
is available at
NEJM.org**



indicating better flow) without angiographic evidence of thrombus.

Key exclusion criteria included ST-segment elevation myocardial infarction within the previous 72 hours, cardiogenic shock or hemodynamic instability, left main coronary artery disease, severe valvular heart disease, a life expectancy of less than 1 year, aorto-ostial lesions with a diameter stenosis of greater than 50%, severe vessel tortuosity, contraindications to the use of adenosine, and lesions located at or supplied by a bypass graft. A complete list of inclusion and exclusion criteria is provided in the Supplementary Appendix.

RANDOMIZATION, TREATMENT, AND FOLLOW-UP

Patients were randomly assigned in a 1:1 ratio to undergo either vFFR-guided or FFR-guided revascularization. Randomization was performed with the use of a concealed, Web-based system, with stratification according to participating center. Patients underwent randomization after clinical and angiographic eligibility criteria were confirmed. Suitable patients provided informed consent before the trial procedures began. The location of the qualifying intermediate lesions (hereafter referred to as study lesions) was captured at baseline.

According to the protocol, physiological assessment of all study lesions was performed with the assigned method (vFFR or FFR) after the administration of a bolus of intracoronary nitrates. The vFFR value was computed on the basis of two angiographic projections acquired at least 30° apart, with the aortic-root pressure measured invasively at the tip of the guiding catheter. Local analysts who had completed the vendor's training and certification process in vFFR analysis performed vFFR directly on site; additional information is provided in the Supplementary Appendix. The FFR was measured with a coronary pressure wire according to standard practice, and hyperemia was induced with intravenous or intracoronary adenosine.

According to the protocol, all study lesions with a measured angiography-based vFFR or pressure-wire-based FFR value of 0.80 or less were to undergo revascularization; if the measurement was more than 0.80, revascularization was to be deferred. Concurrent clinically significant lesions (diameter stenosis of >80%) and culprit lesions

in patients with myocardial infarction underwent revascularization according to routine clinical practice.

Percutaneous revascularization and the use of antithrombotic therapy was performed in accordance with applicable revascularization guidelines of the European Society of Cardiology.^{2,3} Invasive imaging was performed at the discretion of the operator. Revascularization with coronary-artery bypass surgery was permitted if deemed to be necessary according to the consensus of the heart team. Staged procedures were allowed and were recommended to be completed within 45 days after the initial revascularization.¹³ Patients and medical staff were aware of the trial-group assignments. Clinical follow-up was performed by telephone or outpatient visit at 30 days and 1 year.

TRIAL END POINTS

The primary end point was a composite of death from any cause, any myocardial infarction, or any revascularization within 1 year after randomization. The secondary end points included study-vessel failure (study vessels were defined as any vessel containing at least one study lesion, and study-vessel failure was defined as death from cardiac causes, study-vessel myocardial infarction, or clinically indicated study-vessel revascularization), individual components of the composite primary end point, stroke, and definite or probable stent thrombosis. Quality of life was assessed with the Seattle Angina Questionnaire (scores range from 0 to 100, with higher scores indicating better health status). Myocardial infarction was defined according to the Fourth Universal Definition for non-procedure-related events and the Academic Research Consortium-2 criteria for procedural events.^{14,15} Data on procedural end points were collected for all the patients who underwent revascularization procedures; repeat revascularizations were classified as clinically indicated if prespecified criteria were met (see the Supplementary Appendix). Details regarding definitions and additional end points are provided in the Supplementary Appendix.

STATISTICAL ANALYSIS

We estimated that a primary end-point event (death from any cause, any myocardial infarction, or any revascularization) would occur in 6.5% of the patients in each group at 1 year.^{16,17} We determined

that a sample of 2122 patients (1061 patients in each group) would provide 80% power at a one-sided alpha level of 2.5% to show that revascularization guided by vFFR was noninferior to revascularization guided by FFR with respect to the primary end point. We determined that the enrollment of 2122 patients with complete data would result in a primary end-point event occurring in 138 patients within 1 year. To account for an estimated 5% of patients discontinuing the trial, we planned to recruit a total of 2228 patients.

Noninferiority would be shown if the upper limit of the (two-sided) 95% confidence interval for the between-group difference in the cumulative incidence of a primary end-point event at 1 year was below a prespecified margin of 3.0 percentage points, equivalent to one-sided noninferiority testing with an alpha of 0.025. The noninferiority margin was selected on the basis of margins used in previous randomized trials that compared alternative physiology-guided revascularization strategies with FFR.^{16,17} The selected noninferiority margin is smaller than the treatment benefit observed in a previous dedicated randomized, controlled trial, which showed that death, nonfatal myocardial infarction, or repeat revascularization occurred in 18.3% of the patients who had undergone percutaneous coronary intervention (PCI) guided by FFR and in 13.2% of the patients who had undergone PCI guided by angiography.¹⁸

Formal testing was conducted only for the primary end point. The analysis used a modified intention-to-treat approach and was performed in the full analysis set, which excluded patients who had undergone randomization but for whom angiography-based vFFR or pressure-wire-based FFR assessment had never been attempted. No interim analyses were performed.

The cumulative incidence at 1 year was calculated with the Kaplan–Meier method. Follow-up data were censored at 1 year after randomization or at the last known event-free time point, whichever came first. The between-group difference in the 1-year cumulative incidence of a primary-end point event, the 95% confidence interval, and the one-sided P value for noninferiority were calculated with the use of the Com-Nougue method.¹⁹ For other end points, the widths of the confidence intervals have not been adjusted for mul-

tiplicity and should not be used to infer definitive treatment effects. Because the statistical analysis plan (available with the protocol) did not include a provision for multiplicity testing of the secondary end points or in the subgroup analyses, the results are reported as point estimates and 95% confidence intervals.

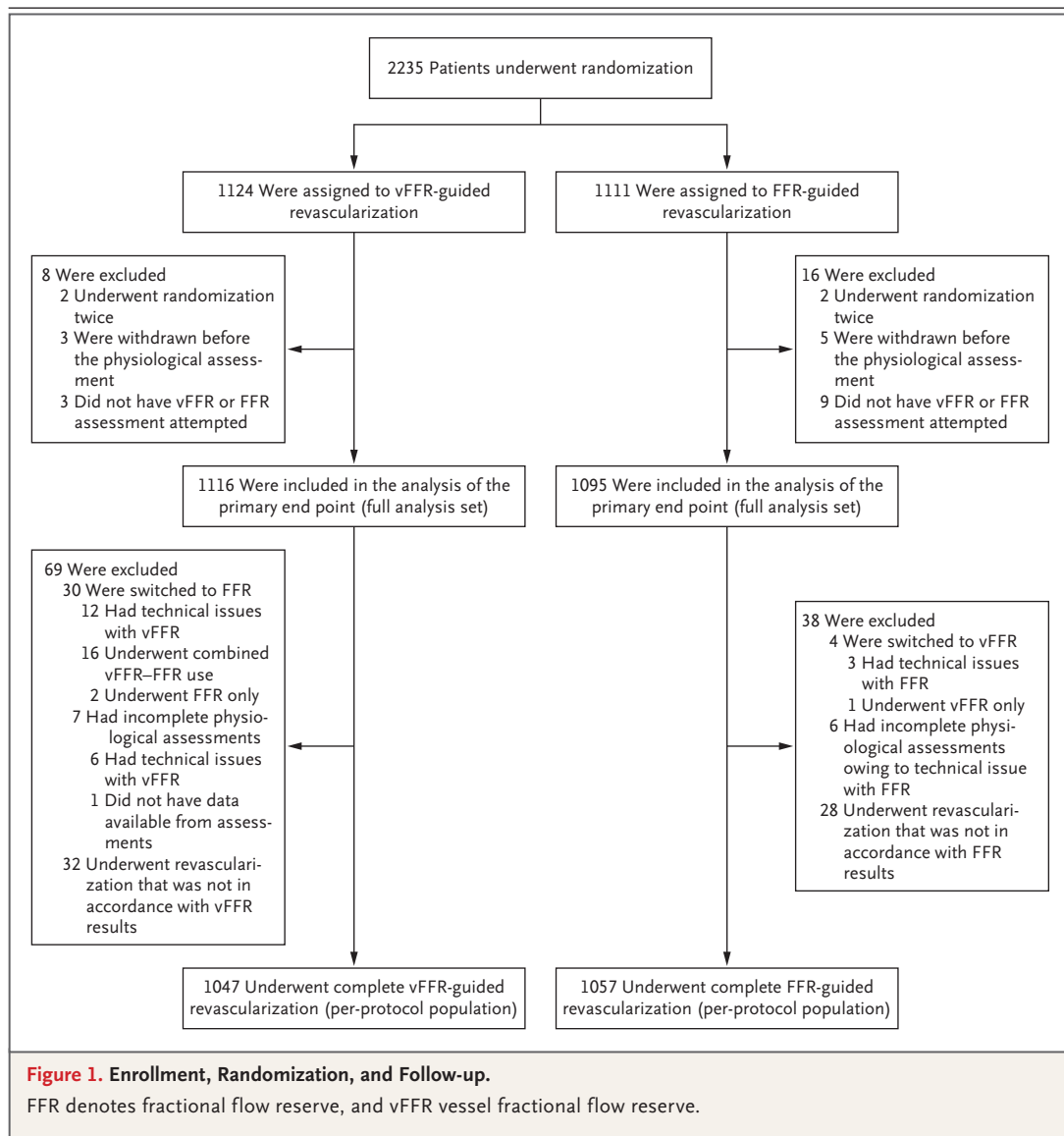
Supportive analyses of the primary end point were conducted in the intention-to-treat population, which included all the patients who had undergone randomization, as well as in the per-protocol population, which included all the patients who had undergone randomization and underwent the assigned physiological assessment of all intermediate lesions, followed by revascularization in accordance with the vFFR or FFR values obtained from the assessment — specifically, revascularization of lesions with a vFFR or FFR value of 0.80 or less and deferral of revascularization of lesions with a vFFR or FFR value of greater than 0.80. Prespecified subgroup analyses were performed for the primary end point in the full analysis set. For secondary end points, including death from cardiovascular causes, competing risks were addressed by censoring follow-up data at the time of the competing event. Missing values were assumed to be missing completely at random, and no imputation or sensitivity analyses were performed. Additional details regarding the statistical analysis are provided in the Supplementary Appendix.

RESULTS

PATIENTS

From November 2021 through May 2024, a total of 2235 patients at 37 centers in seven European countries underwent randomization. The full analysis set included 1116 patients assigned to vFFR-guided revascularization and 1095 assigned to FFR-guided revascularization (Fig. 1). The two groups appeared to be well balanced with respect to baseline clinical and angiographic characteristics (Table 1 and Table S1 in the Supplementary Appendix) and appeared to be representative of the population of patients undergoing PCI for intermediate coronary-artery lesions in modern European practice (Table S2).

The mean age of the patients was 67 years; 24.3% were women, 18.7% presented with an acute coronary syndrome, and 26.6% had diabetes



mellitus. The mean ($\pm$ SD) number of study lesions per patient was 1.27 ± 0.55 in the vFFR group and 1.28 ± 0.55 in the FFR group. The location of the study lesions appeared to be similar in the two groups.

PROCEDURAL CHARACTERISTICS

The characteristics of the procedures are shown in Table 2. Complete physiological assessment of lesions was successful according to the assigned strategy in 96.7% of the patients in the vFFR group and in 99.1% of those in the FFR group. The median vFFR value was 0.83 (interquartile range, 0.74 to 0.89), and the median FFR value was

0.85 (interquartile range, 0.79 to 0.91). Frequency histograms are shown in Figure S1. Functional physiological significance (a vFFR or FFR value of ≤ 0.80) was observed in 40.9% of the study lesions in the vFFR group and in 31.3% of the lesions in the FFR group. The percentages of functionally significant study lesions according to coronary-artery location are shown in Table S3.

Revascularization of study lesions was performed in 45.0% of the patients in the vFFR group and in 36.0% of those in the FFR group. The total number of stents per patient and total length of the stents are shown in Table 2 and Table S4. Among the patients who underwent revascularization of

Table 1. Characteristics of the Patients at Baseline (Full Analysis Set).*

Characteristic	vFFR-Guided Revascularization (N = 1116)	FFR-Guided Revascularization (N = 1095)
Age — yr	67.6±9.9	67.1±9.7
Female sex — no. (%)	266 (23.8)	272 (24.8)
Clinical presentation — no. (%)		
Chronic coronary syndrome	911 (81.6)	886 (80.9)
Unstable angina	77 (6.9)	75 (6.8)
Non-STEMI†	128 (11.5)	134 (12.2)
Medical history — no. (%)		
Hypertension	808 (72.4)	794 (72.5)
Diabetes mellitus	308 (27.6)	281 (25.7)
Dyslipidemia	793 (71.1)	784 (71.6)
Current smoker	221 (19.8)	195 (17.8)
History of PCI	391 (35.0)	380 (34.7)
History of CABG	8 (0.7)	9 (0.8)
History of myocardial infarction	264 (23.7)	251 (22.9)
Peripheral vascular disease	76 (6.8)	97 (8.9)
Atrial fibrillation	147 (13.2)	152 (13.9)
Left ventricular ejection fraction — %‡	54.1±9.9	54.6±10.2
Body-mass index§	27.8±4.8	27.6±4.7
Median eGFR (IQR) — ml/min/1.73 m²¶	76 (56–97)	74 (56–94)
Anatomic SYNTAX Score		
Mean	11.8±8.2	11.5±7.7
Score of <11 — no./total no. (%)	597/1029 (58.0)	604/1031 (58.6)
Score of 11–22 — no./total no. (%)	308/1029 (29.9)	323/1031 (31.3)
Score of >22 — no./total no. (%)	124/1029 (12.1)	104/1031 (10.1)

* Plus-minus values are means ±SD. The full analysis set excluded patients who had undergone randomization but for whom angiography-based vFFR or pressure-wire-based FFR assessment had never been attempted. CABG denotes coronary-artery bypass grafting, eGFR estimated glomerular filtration rate, FFR fractional flow reserve, IQR interquartile range, PCI percutaneous coronary intervention, and vFFR vessel fractional flow reserve.

† Data are included for two patients (one in each group) who presented with ST-segment elevation myocardial infarction (STEMI); both patients were treated after stabilization of the acute presentation in a nonculprit territory. The event in the vFFR group was considered to be a protocol violation.

‡ Data were available for 1210 patients (603 in the vFFR group and 607 in the FFR group).

§ The body-mass index is the weight in kilograms divided by the square of the height in meters.

¶ Data were available for 2137 patients (1074 in the vFFR group and 1063 in the FFR group).

|| The Anatomic SYNTAX Score is an angiographic grading tool used to determine the complexity of coronary artery disease; the lowest score is 0, and higher scores (no upper limit) indicate more complex coronary anatomy. The scores were calculated centrally at the core trial laboratory.

study lesions, revascularization was performed in accordance with the outcome of the physiological assessment in 96.0% of the patients in the vFFR group and in 96.2% in the FFR group.

Among the patients who underwent PCI, the mean (±SD) procedure time was 55.8±26.8 minutes in the vFFR group and 60.9±28.5 minutes in the FFR group (difference, −5.13 minutes; 95%

confidence interval [CI], −8.55 to −1.71). Additional data are provided in Table S5.

END POINTS

A total of 97.1% of the patients in the vFFR group and 96.7% of those in the FFR group completed the 1-year follow-up. In the full analysis set, a primary end-point event occurred in 80 patients

Table 2. Procedural Characteristics (Full Analysis Set).*

Variable	vFFR-Guided Revascularization (N=1116)	FFR-Guided Revascularization (N=1095)
Physiological assessment — no. (%)†		
Complete assessment	1079 (96.7)	1085 (99.1)
Assessment completed with the non-randomly assigned strategy	30 (2.7)	4 (0.4)
Incomplete assessment	7 (0.6)	6 (0.5)
Outcome of physiological assessment — no. (%)		
vFFR or FFR value of ≤ 0.80 in ≥ 1 study lesion‡§	510 (45.7)	401 (36.6)
vFFR or FFR value of > 0.80 in all study lesions‡§	601 (53.9)	688 (62.8)
Assessment could not be performed because of missing vFFR or FFR measurements	5 (0.4)	6 (0.5)
Revascularization		
Revascularization of ≥ 1 study lesion performed — no. (%)	502 (45.0)	394 (36.0)
PCI — no./total no. (%)¶	481/502 (95.8)	373/394 (94.7)
CABG — no./total no. (%)¶	21/502 (4.2)	21/394 (5.3)
Revascularization of all study lesions performed in accordance with outcome of physiological assessment — no./total no. (%)‡	482/502 (96.0)	379/394 (96.2)
Revascularization of ≥ 1 study lesion not performed in accordance with outcome of physiological assessment — no./total no. (%)‡§	20/502 (4.0)	15/394 (3.8)
Deferral of revascularization of all study lesions — no. (%)	614 (55.0)	701 (64.0)
Deferral of revascularization of all study lesions in accordance with outcome of physiological assessment — no./total no. (%)‡	592/614 (96.4)	682/701 (97.3)
Deferral of revascularization of ≥ 1 study lesion not in accordance with outcome of physiological assessment — no./total no. (%)‡§	22/614 (3.6)	19/701 (2.7)
Any revascularization performed — no. (%)¶	659 (59.1)	568 (51.9)
PCI — no./total no. (%)	637/659 (96.7)	547/568 (96.3)
CABG — no./total no. (%)	22/659 (3.3)	21/568 (3.7)
No. of stents placed per patient	0.92 $\pm$ 1.13	0.80 $\pm$ 1.06
No. among patients who underwent PCI	1.60 $\pm$ 1.07	1.60 $\pm$ 0.99
Stent length per patient — mm	23.1 $\pm$ 30.6	20.7 $\pm$ 30.0
Use of intravascular imaging — no. (%)	120 (10.8)	99 (9.0)
Physiological assessment — no. of lesions/total no. (%)		
Assessment successful according to random assignment	1383/1417 (97.6)	1391/1403 (99.1)
Assessment successful with non-randomly assigned strategy	27/1417 (1.9)	6/1403 (0.4)
Assessment not performed successfully	7/1417 (0.5)	6/1403 (0.4)
Outcome of physiological assessment		
Mean vFFR or FFR value	0.80 $\pm$ 0.12	0.84 $\pm$ 0.09
Median vFFR or FFR value (IQR)	0.83 (0.74–0.89)	0.85 (0.79–0.91)
vFFR or FFR value ≤ 0.80 — no. of lesions/total no. (%)‡	580/1417 (40.9)	439/1403 (31.3)
vFFR or FFR value > 0.80 — no. of lesions/total no. (%)‡	830/1417 (58.6)	958/1403 (68.3)
vFFR or FFR value not available — no. of lesions/total no. (%)‡	7/1417 (0.5)	6/1403 (0.4)

* Plus-minus values are means $\pm$ SD. Study lesions were defined as intermediate lesions (diameter stenosis of 30 to 80%) that were identified at baseline and used to determine trial eligibility.

† Complete assessment was defined as assessment that was successfully performed in all study lesions according to the randomly assigned guidance strategy. Assessments completed with the non-randomly assigned strategy refer to complete assessments that were performed with FFR among patients in the vFFR group and complete assessments that were performed with vFFR among patients in the FFR group. Physiological assessment (with either strategy) was considered to be incomplete if the assessment was not successfully performed in at least one study lesion.

‡ For lesions with multiple assessments, the results from the physiological assessment with the assigned strategy were prioritized.

§ Data are included for patients who had multiple study lesions with partially missing data from physiological assessments.

¶ One patient in the vFFR group underwent both PCI and CABG and is counted in the CABG category.

|| One patient in the FFR group underwent both PCI and CABG and is counted in the CABG category.

(Kaplan–Meier estimate, 7.5%) in the vFFR group and in 79 patients (Kaplan–Meier estimate, 7.5%) in the FFR group (risk difference, -0.02 percentage points; 95% CI, -2.25 to 2.21 ; $P=0.004$ for noninferiority) (Table 3 and Fig. 2). Study-vessel failure occurred in 43 patients (Kaplan–Meier estimate, 4.0%) in the vFFR group and in 49 (Kaplan–Meier estimate, 4.6%) in the FFR group (risk difference, -0.62 percentage points; 95% CI, -2.35 to 1.10). Sensitivity analyses in the per-protocol population and in the intention-to-treat population yielded similar results (Table 3 and Table S6). Death from any cause occurred in 2.2% of the patients in the vFFR group and in 2.3% in the FFR group; any myocardial infarction occurred in 2.9% and 2.4%, respectively; and any revascularization occurred in 4.2% in each group. Other secondary end points appeared to be similar in the two groups (Table 3). The results of prespecified subgroup analyses of the primary end point were similar to those of the main analysis (Fig. S2). The mean ($\pm$ SD) Seattle Angina Questionnaire summary score at 30 days was 84.4 ± 16.4 in the vFFR group and 84.3 ± 16.5 in the FFR group, and the score at 1 year was 87.3 ± 14.4 and 86.8 ± 15.1 , respectively (Table S7).

SAFETY

A complete list of site-reported serious adverse events, grouped according to system organ class, is provided in Table S8. The results were similar in the two groups. Intraprocedural complications related to percutaneous revascularization at the time of physiological assessment occurred in 3.7% of the patients in the vFFR group and in 6.0% of those in the FFR group (Table S9).

DISCUSSION

In this international, randomized trial involving patients with intermediate coronary-artery lesions and an indication for physiological assessment, a revascularization strategy guided by vFFR was noninferior to a strategy guided by FFR with respect to a composite of death from any cause, any myocardial infarction, or any revascularization at 1 year. The incidence of study-vessel failure (a composite of death from cardiac causes, study-vessel myocardial infarction, or clinically indicated study-vessel revascularization) appeared to be similar in the two groups. The results of pre-

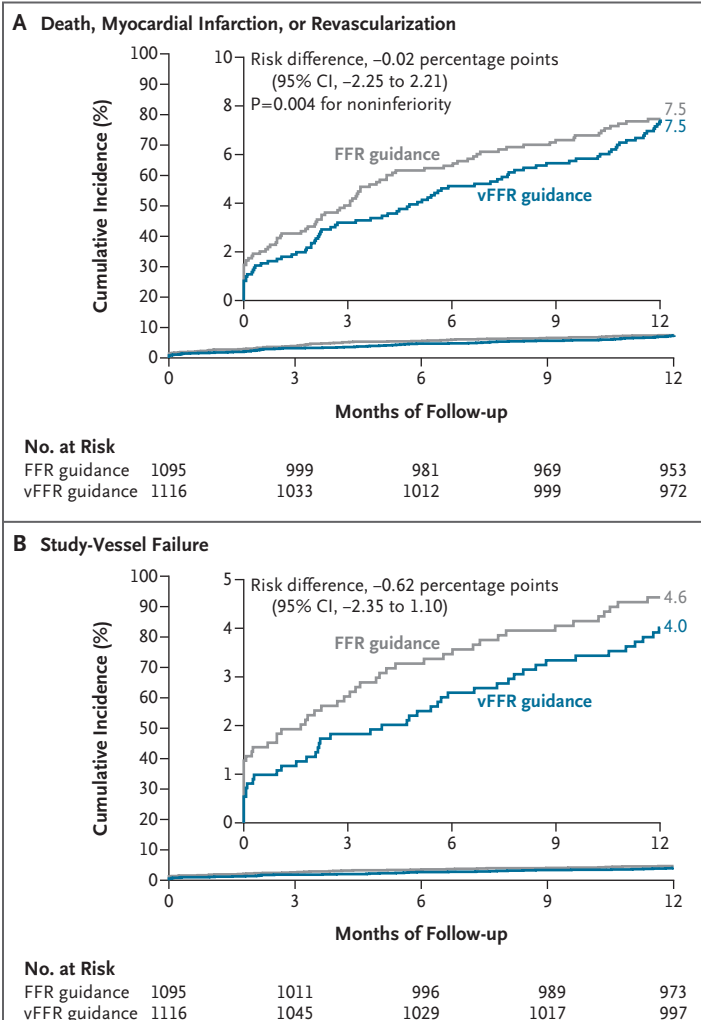


Figure 2. Time-to-Event Curves for the Primary and Secondary End Points.

Panel A shows the cumulative incidence of a primary composite end-point event (death from any cause, any myocardial infarction, or any revascularization). Panel B shows the incidence of study-vessel failure, defined as a composite of death from cardiac causes, study-vessel myocardial infarction, or clinically indicated study-vessel revascularization. The inset in each panel shows the same data on an enlarged y axis. Study vessels were defined as any vessel containing at least one intermediate lesion (diameter stenosis of 30 to 80%) that was identified at baseline and used to determine trial eligibility. The widths of the confidence intervals in Panel B have not been adjusted for multiplicity and cannot be used to infer treatment effects.

specified per-protocol and subgroup analyses appeared to be similar to those of the main analyses.

A higher percentage of functionally significant lesions were identified with vFFR than with FFR, resulting in a between-group difference of 9 percentage points in the percentage of patients

Table 3. Primary and Secondary End Points.*

End Point	Full Analysis Set			Per-Protocol Population		
	vFFR-Guided Revascularization (N=1116)	FFR-Guided Revascularization (N=1095)	Risk Difference (95% CI)†	vFFR-Guided Revascularization (N=1047)	FFR-Guided Revascularization (N=1057)	Risk Difference (95% CI)†
<i>no. (%)</i>						
Primary end point						
Death from any cause, any myocardial infarction, or any revascularization	80 (7.5)	79 (7.5)	-0.02 (-2.25 to 2.21)‡	71 (7.0)	76 (7.4)	-0.40 (-2.65 to 1.86)
Secondary end points						
Study-vessel failure: death from cardiac causes, study-vessel myocardial infarction, or clinically indicated study-vessel revascularization	43 (4.0)	49 (4.6)	-0.62 (-2.35 to 1.10)	38 (3.8)	47 (4.6)	-0.83 (-2.57 to 0.91)
Patient-oriented composite end point§	85 (7.9)	82 (7.8)	0.16 (-2.12 to 2.44)	76 (7.5)	79 (7.7)	-0.20 (-2.51 to 2.11)
Device-oriented composite end point¶	40 (3.7)	41 (3.9)	-0.14 (-1.76 to 1.47)	35 (3.4)	41 (4.0)	-0.54 (-2.18 to 1.10)
Study-oriented composite end point	60 (5.6)	60 (5.7)	-0.08 (-2.04 to 1.87)	52 (5.1)	58 (5.7)	-0.52 (-2.49 to 1.44)
Target-vessel failure**	40 (3.7)	41 (3.9)	-0.14 (-1.76 to 1.48)	36 (3.6)	41 (4.0)	-0.43 (-2.09 to 1.22)
Target-lesion failure***	37 (3.4)	40 (3.8)	-0.33 (-1.90 to 1.25)	33 (3.3)	40 (3.9)	-0.63 (-2.25 to 0.98)
Death from any cause	23 (2.2)	24 (2.3)	-0.11 (-1.37 to 1.14)	19 (1.9)	23 (2.3)	-0.36 (-1.60 to 0.89)
Death from cardiac causes	11 (1.0)	16 (1.5)	-0.47 (-1.43 to 0.48)	9 (0.9)	16 (1.6)	-0.66 (-1.63 to 0.30)
Any myocardial infarction††	31 (2.9)	26 (2.4)	0.44 (-0.92 to 1.80)	28 (2.8)	26 (2.5)	0.24 (-1.15 to 1.64)
Periprocedural myocardial infarction	11 (1.0)	13 (1.2)	-0.20 (-1.08 to 0.68)	10 (1.0)	13 (1.2)	-0.27 (-1.17 to 0.62)
Study-vessel myocardial infarction	19 (1.8)	19 (1.8)	-0.02 (-1.13 to 1.10)	16 (1.6)	19 (1.8)	-0.26 (-1.38 to 0.86)
Any revascularization‡‡	45 (4.2)	44 (4.2)	0.03 (-1.69 to 1.75)	40 (4.0)	42 (4.2)	-0.16 (-1.89 to 1.57)
Clinically indicated revascularization	38 (3.6)	37 (3.5)	0.04 (-1.52 to 1.62)	34 (3.4)	35 (3.5)	-0.06 (-1.66 to 1.53)
Clinically indicated study-vessel revascularization	22 (2.1)	24 (2.3)	-0.23 (-1.48 to 1.02)	19 (1.9)	22 (2.2)	-0.28 (-1.52 to 0.96)
Any stroke	7 (0.6)	4 (0.4)	0.27 (-0.33 to 0.87)	7 (0.7)	4 (0.4)	0.30 (-0.33 to 0.93)
Definite or probable stent thrombosis§§	2 (0.2)	3 (0.3)	-0.09 (-0.49 to 0.32)	2 (0.2)	3 (0.3)	-0.09 (-0.51 to 0.34)

* All the end points are reported at 1 year of follow-up. The reported percentages are Kaplan–Meier estimates at 1 year. The widths of the confidence intervals have not been adjusted for multiplicity and cannot be used to infer treatment effects. The per-protocol population included all the patients who had undergone randomization, had successful assessment of all intermediate lesions according to the assigned strategy, and underwent revascularization on the basis of the vFFR or FFR values obtained from the assessment. Target lesions were defined as those that underwent revascularization during the index procedure, irrespective of their categorization as study lesions. Target vessels were defined as any vessel containing at least one target lesion.

† The values are the differences in percentage points.

‡ $P=0.004$ for noninferiority.

§ The patient-oriented composite end point was death from any cause, any stroke, any myocardial infarction, or any revascularization.

|| The device-oriented composite end point was death from cardiovascular causes, target-vessel myocardial infarction, or clinically indicated repeat revascularization of the target lesion.

** The study-oriented composite end point was death from cardiovascular causes, study-vessel or target-vessel myocardial infarction, or study-vessel or target-vessel revascularization. Target-vessel failure was defined as a composite of death from cardiac causes, target-vessel myocardial infarction, or clinically indicated target-vessel revascularization.

†† Failure was defined as a composite of death from cardiac causes, target-vessel myocardial infarction, or clinically indicated target-lesion revascularization.

‡‡ Non-procedure-related myocardial infarction was defined according to the Fourth Universal Definition of Myocardial Infarction. Procedure-related myocardial infarction was defined according to the Academic Research Consortium–2 criteria.

§§ Any revascularization included all first clinically indicated elective, urgent, or emergency revascularization procedures that were not planned during the index hospitalization during the overall trial period.

||| Stent thrombosis was defined according to the Academic Research Consortium–2 criteria.

who underwent revascularization of study lesions; this finding is consistent with a recent trial comparing quantitative flow reserve with FFR-guided revascularization.¹¹ Although this finding was not readily apparent in initial validation studies, several aspects have been hypothesized to contribute to the discordance in the results from angiography-based FFR and pressure-wire–measured FFR. Angiography-based physiological assessment often assumes higher flow than actually occurs because it does not take into account actual flow demand, microvascular resistance, and patient-specific response to hyperemia, factors that can result in increased pressure drops and lower physiological values.^{20–22} Furthermore, because of accumulated resistance along the vessel, angiography-based physiological assessment of longer or tapered lesions may overestimate the severity of the lesions in the case of diffuse disease.²³ Future study is warranted to determine whether the higher percentage of patients in the vFFR group than in the FFR group who underwent revascularization indicates that vFFR better detected physiologically relevant lesions or whether there was a systemic shift in decision thresholds on the basis of patient characteristics and lesion location. A prespecified cost-effectiveness analysis is planned to determine the implications of the increased number of revascularizations at baseline with vFFR given the shorter procedures without the need for pressure wires or adenosine.

Despite the above-mentioned differences between vFFR and FFR, a vFFR-guided revascularization strategy, with subsequent higher revascularization rates than a revascularization strategy guided by pressure-wire–based FFR, resulted in similar outcomes at 1 year. This finding is consistent with the results of the FLAVOUR (Fractional Flow Reserve and Intravascular Ultrasound-Guided Intervention Strategy for Clinical Outcomes in Patients with Intermediate Stenosis) trial, which compared FFR with intravascular-ultrasonography–guided revascularization. In the FLAVOUR trial, although more patients in the intravascular ultrasonography–guided group than in the angiography-guided group underwent revascularization (difference, 21 percentage points), 2-year clinical outcomes were similar in the two groups.²⁴ Future studies are needed to establish the respective role of imaging as compared with physiological assessment of intermediate coronary lesions in patients with both stable and unstable coronary artery disease.²⁵

With the results of numerous clinical validation studies supporting a high diagnostic accuracy, the findings in the current trial support the use of an angiography-based method that does not require the use of a pressure wire or induced hyperemia when physiological guidance of revascularization is indicated. The results of this trial contribute to the knowledge base of physiology-guided revascularization in showing that angiography-based vFFR was noninferior to pressure-wire-based FFR in patients with chronic and acute coronary syndromes and intermediate coronary-artery lesions.

Our trial has several limitations. First, the open-label design could have introduced bias in clinical decision making after the initial physiological assessment; however, all the end-point events were adjudicated by an independent clinical-events committee. Second, the number of patients who had an acute coronary syndrome was small, which limits generalizability to this patient population. Third, although investigators were trained in the use of vFFR before enrollment began, experience with pressure-wire-based FFR was far more extensive, which may have favored the FFR group. Fourth, the trial was conducted in high-volume European centers that had experience in coronary-artery physiological assessment; results may not be generalizable to lower-volume centers or regions with less experience in angiography-based computational tools. Moreover, data are representative primarily of patients in southern Europe and the Netherlands and may not be generalizable to other populations or health care systems. Finally, follow-up was limited to 1 year; longer follow-up could have provided valuable information on the long-term effects of the events that occurred during the first year.

Among patients with intermediate coronary-artery lesions, a vFFR-guided revascularization strategy was noninferior to an FFR-guided strategy with respect to a composite of death, myocardial infarction, or revascularization at 1 year.

Supported by unrestricted grants from Pie Medical Imaging and Siemens Healthineers to the European Cardiovascular Research Institute.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

AUTHOR INFORMATION

¹Department of Cardiology, Thoraxcenter, Erasmus University Medical Center, Rotterdam, the Netherlands; ²Department of Cardiology, Ospedale dell'Angelo—ULSS 3 Serenissima, Mestre, Italy; ³Department of Cardiology, Mater Private Network, Dublin; ⁴Cardiovascular Research Institute Dublin, Mater Private Network, Dublin; ⁵School of Pharmacy and Biomolecular Sciences, Royal College of Surgeons in Ireland University of Medicine and Health Sciences, Dublin; ⁶Department of Cardiology, Ospedale Papa Giovanni XXIII, Bergamo, Italy; ⁷Oxford Heart Centre, John Radcliffe, Oxford University Hospitals, National Health Service Foundation Trust, Oxford, United Kingdom; ⁸Department of Cardiology, Institut Coeur Poumon, Centre Hospitalier Universitaire Lille, Lille, France; ⁹Department of Interventional Cardiology for Coronary Valves and Structural Heart Diseases, INSERM, U1011, Institut Pasteur de Lille, European Genomic Institute for Diabetes, Université de Lille, Lille, France; ¹⁰Department of Cardiology, Hospital Clínico Universitario Valladolid, Valladolid, Spain; ¹¹Department of Cardiology, IRCCS Humanitas Research Hospital, Milan; ¹²Department of Cardiology, Institut Cardiovasculaire Paris Sud, Hôpital Jacques Cartier, Ramsay Santé, Massy, France; ¹³Cardiology Department, Cardiovascular Institute, Hospital Clinic, Institut d'Investigacions Biomèdiques August Pi i Sunyer, University of Barcelona, Centro de Investigación Biomédica en Red Enfermedades Cardiovasculares (CIBER-CV), Barcelona; ¹⁴Department of Cardiology, Mater Private Cork, Cork, Ireland; ¹⁵Department of Cardiology, Azienda Ospedaliero Universitaria di Ferrara, Ferrara, Italy; ¹⁶Department of Cardiology, Instituto de Investigación Sanitaria Hospital Universitario de La Princesa, Universidad Autónoma de Madrid, CIBER-CV, Madrid; ¹⁷Dipartimento di Medicina Transazionale, Università del Piemonte Orientale, Novara, Italy; ¹⁸Department of Cardiology, Ospedale Maggiore della Carità, Novara, Italy; ¹⁹Department of Cardiology, Angiology, and Intensive Care Medicine, Deutsches Herzzentrum der Charité, Charité—Universitätsmedizin Berlin, Berlin; ²⁰Deutsches Zentrum für Herz-Kreislauf-Forschung, Partner Site Berlin, Berlin; ²¹Department of Cardiology, Albert Schweitzer Hospital, Dordrecht, the Netherlands; ²²Department of Cardiology, La Paz Hospital, Madrid; ²³Department of Cardiology, Royal Sussex County Hospital, Brighton, United Kingdom; ²⁴Coronary Revascularization Unit, IRCCS Policlinico San Donato, San Donato Milanese, Italy; ²⁵Cardialysis, Rotterdam, the Netherlands; ²⁶National Institute for Health and Care Research Oxford Biomedical Research Centre, John Radcliffe Hospital, Oxford, United Kingdom; ²⁷Amsterdam University Medical Centers, Department of Cardiology, Amsterdam; ²⁸European Cardiovascular Research Institute, Rotterdam, the Netherlands.

REFERENCES

1. Lawton JS, Tamis-Holland JE, Bangalore S, et al. 2021 ACC/AHA/SCAI guideline for coronary artery revascularization: a report of the American College of Cardiology/American Heart Association Joint Committee on clinical practice guidelines. *Circulation* 2022;145(3):e18–e114.
2. Neumann F-J, Sousa-Uva M, Ahlsson A, et al. 2018 ESC/EACTS guidelines on myocardial revascularization. *Eur Heart J* 2019;40:87–165.
3. Vrints C, Andreotti F, Koskinas KC,

- et al. 2024 ESC guidelines for the management of chronic coronary syndromes. *Eur Heart J* 2024;45:3415-537.
4. De Maria GL, Garcia-Garcia HM, Scarsini R, et al. Novel indices of coronary physiology: do we need alternatives to fractional flow reserve? *Circ Cardiovasc Interv* 2020;13(4):e008487.
5. Gould KL, Kelley KO, Bolson EL. Experimental validation of quantitative coronary arteriography for determining pressure-flow characteristics of coronary stenosis. *Circulation* 1982;66:930-7.
6. Kirkeeide RL. Coronary obstructions, morphology and physiologic significance. In: Reiber JHC, Serruys PW, eds. *Quantitative coronary arteriography: developments in cardiovascular medicine*. Vol. 117. Dordrecht, the Netherlands: Springer Science+Business Media, 1991:229-44.
7. Masdjedi K, Tanaka N, Van Belle E, et al. Vessel fractional flow reserve (vFFR) for the assessment of stenosis severity: the FAST II study. *EuroIntervention* 2022;17:1498-505.
8. Masdjedi K, van Zandvoort LJC, Balbi MM, et al. Validation of a three-dimensional quantitative coronary angiography-based software to calculate fractional flow reserve: the FAST study. *EuroIntervention* 2020;16:591-9.
9. Neleman T, Masdjedi K, Van Zandvoort LJC, et al. Extended validation of novel 3D quantitative coronary angiography-based software to calculate vFFR: the FAST EXTEND study. *JACC Cardiovasc Imaging* 2021;14:504-6.
10. Xu B, Tu S, Song L, et al. Angiographic quantitative flow ratio-guided coronary intervention (FAVOR III China): a multicentre, randomised, sham-controlled trial. *Lancet* 2021;398:2149-59.
11. Andersen BK, Sejr-Hansen M, Mailard L, et al. Quantitative flow ratio versus fractional flow reserve for coronary revascularisation guidance (FAVOR III Europe): a multicentre, randomised, non-inferiority trial. *Lancet* 2024;404:1835-46.
12. Scoccia A, Byrne RA, Banning AP, et al. Fractional flow reserve or 3D-quantitative-coronary-angiography based vessel-FFR guided revascularization: rationale and study design of the prospective randomized FAST III trial. *Am Heart J* 2023;260:1-8.
13. Spitzer E, McFadden E, Vranckx P, et al. Defining staged procedures for percutaneous coronary intervention trials: a guidance document. *JACC Cardiovasc Interv* 2018;11:823-32.
14. Thygesen K, Alpert JS, Jaffe AS, et al. Fourth universal definition of myocardial infarction (2018). *J Am Coll Cardiol* 2018;72:2231-64.
15. Garcia-Garcia HM, McFadden EP, Farb A, et al. Standardized end point definitions for coronary intervention trials: the Academic Research Consortium-2 consensus document. *Eur Heart J* 2018;39:2192-207.
16. Götzberg M, Christiansen EH, Gudmundsdottir JJ, et al. Instantaneous wave-free ratio versus fractional flow reserve to guide PCI. *N Engl J Med* 2017;376:1813-23.
17. Davies JE, Sen S, Dehbi H-M, et al. Use of the instantaneous wave-free ratio or fractional flow reserve in PCI. *N Engl J Med* 2017;376:1824-34.
18. Tonino PA, De Bruyne B, Pijls NHJ, et al. Fractional flow reserve versus angiography for guiding percutaneous coronary intervention. *N Engl J Med* 2009;360:213-24.
19. Com-Nougue C, Rodary C, Patte C. How to establish equivalence when data are censored: a randomized trial of treatments for B non-Hodgkin lymphoma. *Stat Med* 1993;12:1353-64.
20. van der Eijk JA, Groenland FTW, Scoccia A, et al. Validation of angiography-based FFR in non-culprit vessels of patients presenting with STEMI. *Clin Res Cardiol* 2025 September 8 (Epub ahead of print).
21. Lee KY, Hwang B-H, Kim MJ, et al. Influence of lesion and disease subsets on the diagnostic performance of the quantitative flow ratio in real-world patients. *Sci Rep* 2021;11:2995.
22. Mejía-Rentería H, Escaned J. Use of fractional flow reserve in non-infarct-related arteries during ST-segment elevation myocardial infarction: the need to move from experimental studies to real life assessment. *JACC Cardiovasc Interv* 2018;11:1659-60.
23. Fezzi S, Del Sole PA, Burzotta F, et al. Angiography-derived physiological patterns of coronary artery disease: implications with post-stenting physiology and long-term clinical outcomes. *Clin Res Cardiol* 2024;113:1745-56.
24. Koo B-K, Hu X, Kang J, et al. Fractional flow reserve or intravascular ultrasonography to guide PCI. *N Engl J Med* 2022;387:779-89.
25. Hu X, Zhang J, Yang S, et al. Angiography-derived fractional flow reserve versus intravascular ultrasound to guide percutaneous coronary intervention in patients with coronary artery disease (FLAVOUR II): a multicentre, randomised, non-inferiority trial. *Lancet* 2025;405:1491-504.

Copyright © 2026 Massachusetts Medical Society.